Counter Mode Using S-DES

23. Write a C program for Encrypt and decrypt in counter mode using one of the following ciphers: affine modulo 256, Hill modulo 256, S-DES. Test data for S-DES using a counter starting at 0000 0000. A binary plaintext of 0000 0001 0000 0010 0000 0100 encrypted with a binary key of 01111 11101 should give a binary plaintext of 0011 1000 0100 1111 0011 0010. Decryption should work correspondingly.

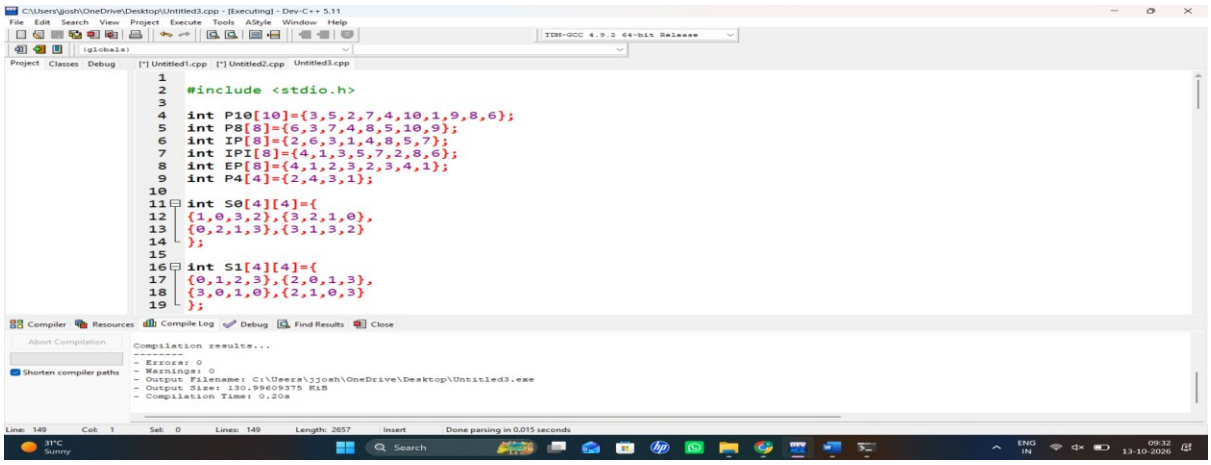
Aim

To write a C program to encrypt and decrypt data using the S-DES algorithm in Counter (CTR) mode, starting with a counter value of 00000000.

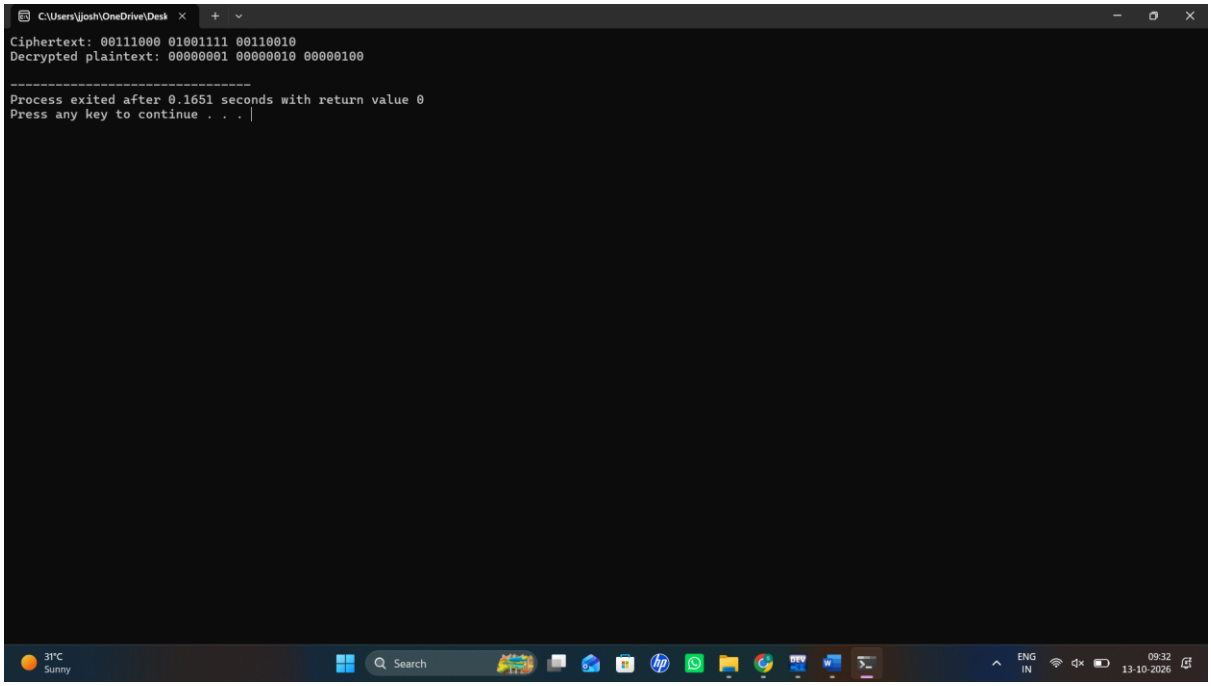
Algorithm

1. Start the program.
2. Read the 10-bit S-DES key and plaintext blocks.
3. Initialize the counter to zero.
4. Encrypt the current counter using S-DES.
5. XOR the encrypted counter with the plaintext block to obtain the ciphertext block.
6. Increment the counter for the next block.
7. For decryption, repeat the same counter encryption and XOR operation with the ciphertext.
8. Display the ciphertext and recovered plaintext.
9. Stop the program.

Program :



Output:



Result

Thus, S-DES encryption and decryption in counter mode were implemented successfully. CTR mode uses the same counter-generated keystream for encryption and decryption.